Subequivariant Graph Reinforcement Learning in 3D Environments

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01 Subequivariant Graph Reinforcement Learning (RL) in **3**D: Title and Authors

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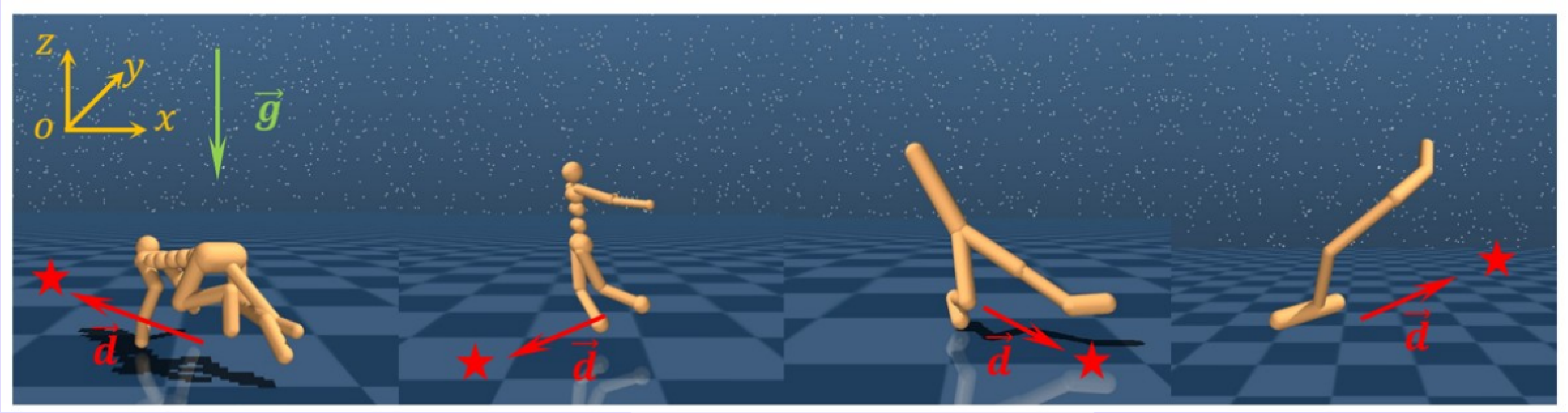
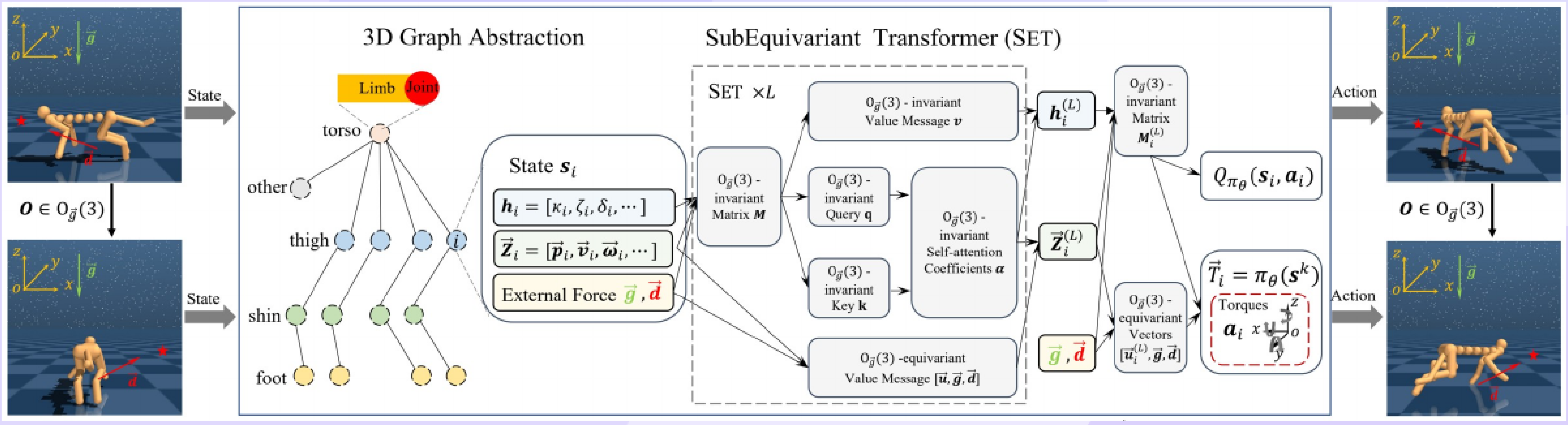
02 Motivation: Beyond **2**D Planar Morphology-Agnostic Reinforcement Learning (RL)



- Move from prior 2D constraints to full 3D with arbitrary orientations and targets.
- Introduce gravity-aware subequivariance to improve optimization and generalization.
- Address exploration difficulty from enlarged state and action spaces.

03 Contributions: Benchmarks, SET, and Results



- Provide 3D Subequivariant Graph Reinforcement Learning (3D-SGRL) benchmarks covering in-domain and cross-domain suites.
- Propose SubEquivariant Transformer (SET) for actor and critic with gravity-aware design.
- Demonstrate consistent gains in single-task, multi-task, and zero-shot settings.

04 Title

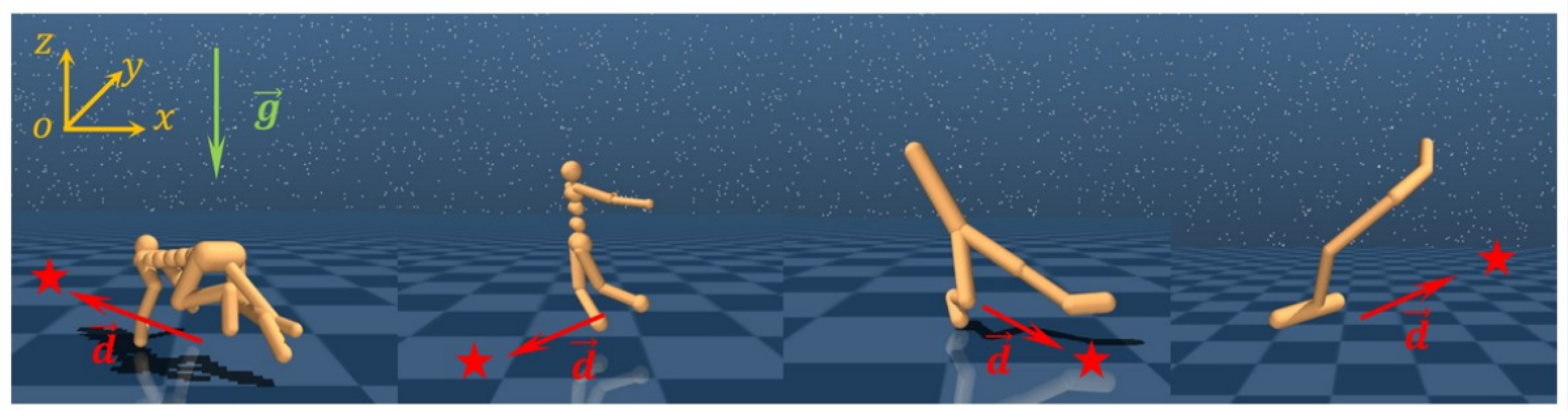
Primer and Geometric Principle

- Represent agents as graphs with shared policies over limbs via Graph Neural Networks (GNNs) or Transformers.
- Train a single actor-critic to control diverse morphologies with weight sharing.
- Leverage message passing to coordinate local limb control and global behavior.

Primer and Geometric Principle

- Model actions that rotate with inputs while preserving the gravity direction.
- Relax full Euclidean symmetry to gravity-preserving **subgroups**.
- Embed gravity and task direction into **message passing** for stability and control.

05 3D-SGRL: Problem Scope and Why 3D Changes the Game



- Full 3D degrees of freedom (DoF) expand observation and control complexity.
- Subequivariant design preserves **useful symmetries** without overconstraining.
- Transformer-based messages remain expressive under **gravity-aware** structure.

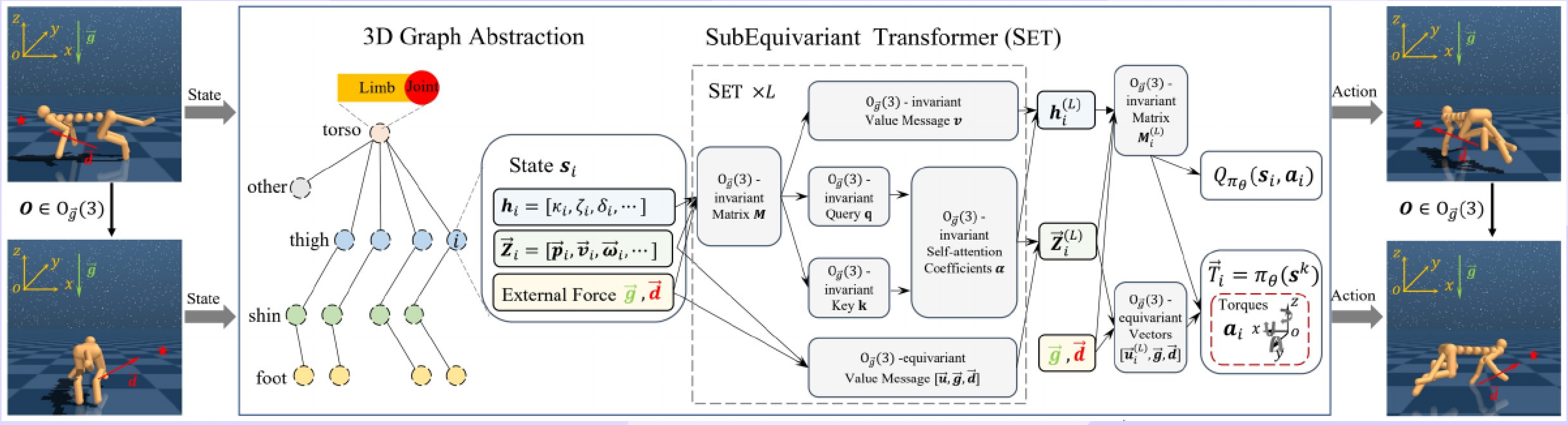
06 Benchmark Suites and Evaluation Roles (High-Level)

- In-domain suites: 3D Hopper++, 3D Walker++, 3D Humanoid++, 3D Cheetah++.
- Cross-domain suites: 3D WHH++ and 3D CWHH++.
- Roles: single-task and multi-task training; zero-shot evaluation.

07 State, Action, and Symmetry Design in 3D-SGRL

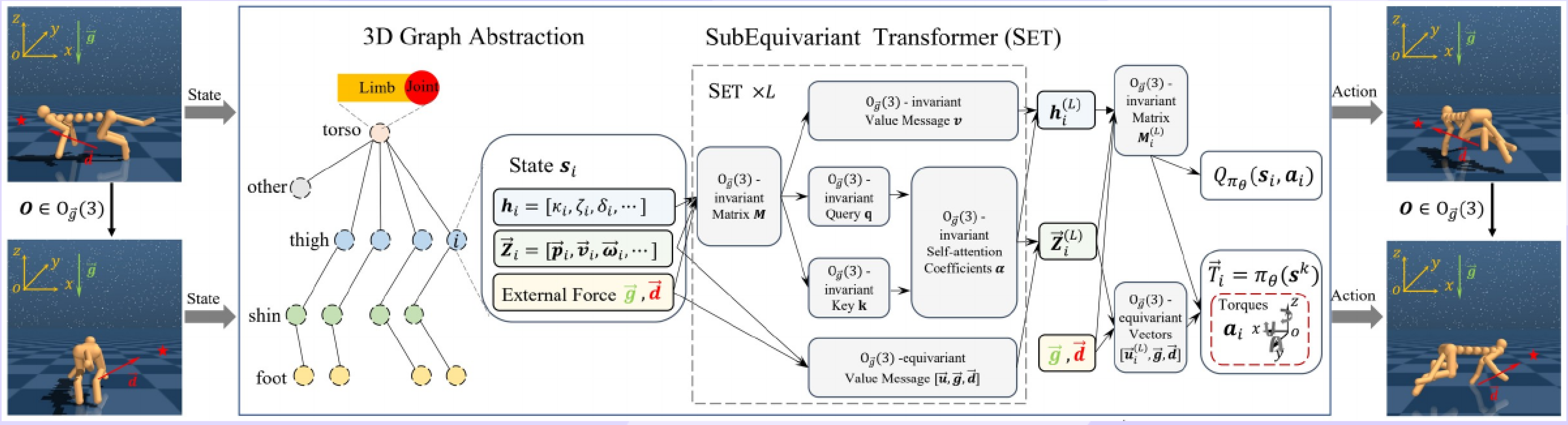
- Separate steerable vectors from invariant scalars to respect geometry.
- Include gravity and task direction to enable subequivariance.
- Project equivariant torques onto joint axes for invariant actuator commands.

08 SubEquivariant Transformer (SET): Message Passing



- Use invariant queries and keys with gravity and task-aware features.
- Aggregate both equivariant and invariant messages for expressivity.
- Maintain subequivariance while normalizing and stacking Transformer layers.

09 Actor (Equivariant) and Critic (Invariant) Designs


3D Graph Abstraction
SubEquivariant Transformer (SET)

- Actor (in Reinforcement Learning, RL) outputs equivariant torques that rotate with inputs.
- Critic computes an invariant value from gravity-aware features.
- Design ensures policy symmetry and value stability across morphologies.

10 Episode Definitions

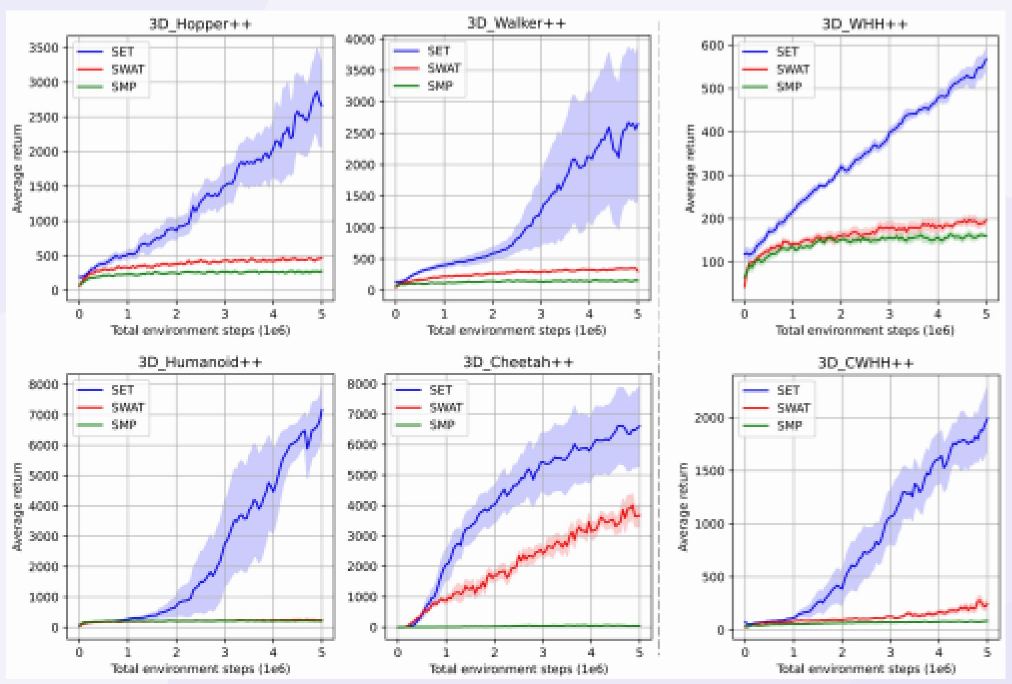
- Episode settings are aligned across suites for fair comparison.
- Initialization covers varied headings to test directional robustness.

11 Evaluation Setup and Baselines

- Baselines: morphology-agnostic SMP and SWAT; monolithic reference.
- Suites: 3D Hopper++, 3D Walker++, 3D Humanoid++, 3D Cheetah++, 3D WHH++, 3D CWHH++.
- Roles: single-task and multi-task training; zero-shot evaluation.

12 Main Results: Multi-Task

- SET surpasses SMP and SWAT across diverse 3D-SGRL tasks.
- Training curves show faster learning and higher returns.
- Baselines struggle under large exploration spaces in 3D.



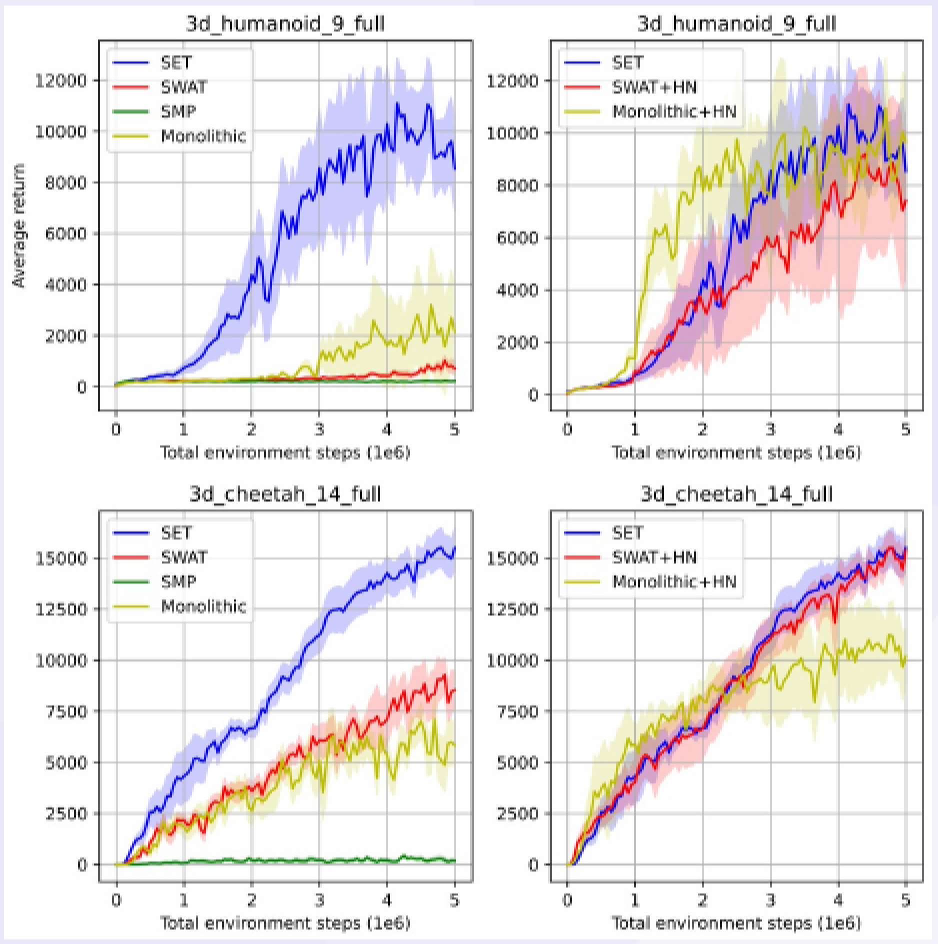
13 Main Results: Zero-Shot and Transfer

- SET outperforms baselines on held-out morphologies in in-domain and cross-domain tests.
- Two-stage transfer on harder variants improves performance; baselines show limited gains.

14 Main Results: Zero-Shot and Transfer (Plots)

- Outperforms baselines on held-out morphologies across suites.
- Two-stage transfer on harder variants improves returns for SET.
- Robust without heading normalization assumptions.

15 Main Results: Single-Task



- SET outperforms Monolithic, SMP, and SWAT in single-task training.
- Transformer expressivity with subequivariance yields robust control.
- Monolithic can win occasionally, but SET maintains advantages.

16 Heading Normalization (HN) vs. SET

- SET matches heading-normalized variants when headings align.
- Under biased forward directions at test time, heading-normalized baselines can fail while SET remains stable.
- Across zero-shot cross-domain tests, SET is competitive with or better than heading-normalized baselines.

17 Title

Ablations and Limitations.

- Gravity and height cues are crucial for biped stability and balance.
- Task direction preservation is essential for goal-reaching performance.
- Equivariant outputs outperform invariant-only readouts.

Limitations

- Fixed gravity assumption may break under changing fields adversely.
- Sensor noise and miscalibration can degrade **equivariant** readouts.
- Transformer message passing increases compute for very high degrees of freedom.

18 Ablations: Key Findings (cont.)

- Removing gravity or height cues degrades biped stability, especially on humanoid tasks.
- Dropping task-direction guidance harms performance across tasks.
- Predicting torques from equivariant outputs is more effective than invariant-only readouts.

19 Open Questions

- Incorporate richer symmetries and handle varying gravity.
- Improve data efficiency with offline learning, demonstrations, or planning.
- Extend to manipulation, multi-robot coordination, and morphology design.

Thank You

Questions & Discussion